

University of Arizona

Crime as it Relates to Income in Tucson: Examining Data-Driven Insights by Ward

Authors

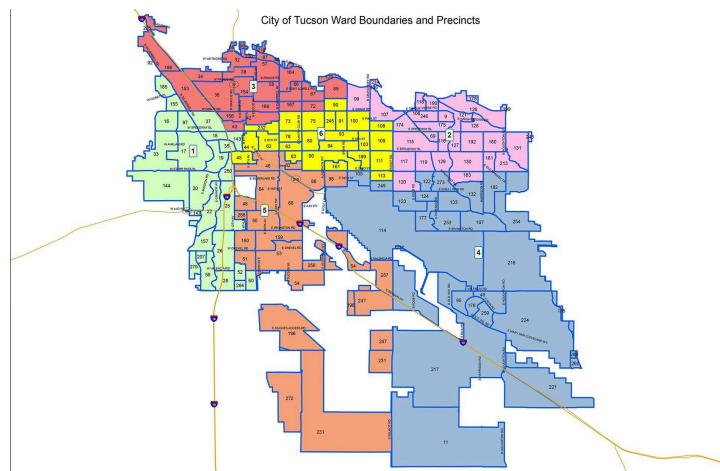
Jon Thrall
Ming Wang
Jaidon McIntosh-Cassa

Class Information

CSC 380 – 001 Principles of Data Science
Professor Cesim Erten
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INTRODUCTION

The quantitative relationship between income levels and crime rates within the wards and neighborhoods of Tucson, Arizona, has been investigated for this report with the goal of addressing the prevailing social problem of economic disparity and its association with criminal activity. The concern is how crime rates differ from one neighborhood to another, and how those differences relate to socioeconomic factors such as income and housing density. We seek to answer the following questions: What is the relationship between income levels and total crime rates across each ward? What temporal patterns emerge in crime across different economic statuses? What are the characteristics of high-crime neighborhoods, and how do they differ from low-crime neighborhoods? Our hypothesis is that lower-income wards are associated with higher levels of crime, and that this relationship can be partially captured through neighborhood-level wealth and household data. We use descriptive statistics, data visualization, and two predictive modeling approaches, linear regression and logistic classification, to explore this relationship and to assess how well socioeconomic indicators alone can explain and predict crime patterns across Tucson's wards.



To left is an image illustrating the wards of Tucson, AZ.

Ward 1 - West ward
(depicted in green)

Ward 2 - North-East ward
(depicted in pink)

Ward 3 - North-West ward
(depicted in red)

Ward 4 - South-East ward
(depicted in blue)

Ward 5 - South-Central ward
(depicted in orange)

Ward 6 - Central-most ward
(depicted in yellow)

Figure 1- <https://news.azpm.org/s/48360-tucson-city-council-two-primary-elections-two-november-elections/>

RELATED WORKS

The following works point to a consistent association between economic conditions and crime rates in Tucson. Ward-level analyses across multiple years converge to show that areas with higher poverty and lower median incomes tend to record higher crime rates, while wealthier, more economically stable wards tend to record lower rates. These findings informed our hypothesis that economic conditions are one factor associated with differences in crime across wards, and motivated the need for continued, careful, evidence-based study rather than firm causal claims.

A 2018 KOLD News 13 report on Ward 3 documented a high volume of violent crime relative to other wards, and linked the pattern to relatively higher joblessness, poverty, and housing instability in the area (Durham). The City of Tucson's 2020 "Poverty and Urban Stress" report similarly found that Wards 3 and 5, whose median incomes fall below the city average, show elevated crime relative to more affluent wards such as 2 and 4. Weimer's analysis of crime determinants in Tucson applies social disorganization theory, finding that economic deprivation and neighborhood instability are associated with variation in violent crime rates across the city. Taken together, these sources support treating income and neighborhood economic stability as relevant variables in understanding crime patterns, while also underscoring that they are only part of a larger, more complex picture that includes policing practices, historical context, and social cohesion — factors outside the scope of the datasets used here.

METHODOLOGY

Outline

This project examines, using data from Tucson, Arizona, the relationship between neighborhood economic characteristics and crime rates. It addresses the following questions:

- (1) What are the temporal patterns of crime across Tucson's wards?
- (2) Is lower neighborhood income associated with higher rates of crime?
- (3) Can neighborhood-level wealth and household data predict the volume of crime in a neighborhood?
- (4) Can neighborhoods be classified as high-crime or low-crime based on these socioeconomic attributes alone?

Data Sources and Integration

The analysis uses datasets obtained from the City of Tucson's public data hub: (1) Crime and arrest records, containing reported incidents by neighborhood, ward, type, and time; and (2) Neighborhood income and household data, containing a wealth index and total household counts at the neighborhood level. These datasets were merged on shared neighborhood identifiers to build a per-neighborhood view combining crime volume with income and household characteristics.

An earlier version of this analysis also incorporated neighborhood-level race and ethnicity demographic data as model features. That data has been removed from the analysis presented here. Race and ethnicity are not variables this project set out to study, and including them as predictive inputs alongside income risked implying a claim this analysis is not equipped to make — that demographic composition itself predicts criminal activity. The revised analysis below uses only income (Wealth Index) and household count as model features, consistent with the project's original hypothesis about economic conditions.

Analytical Framework

The project applies the following approaches: Correlation and visual analysis, to explore the relationship between the Wealth Index and crime counts; Linear Regression, to model the association between neighborhood wealth/household characteristics and total crime count; and Logistic Regression, to classify neighborhoods as high-crime or low-crime based on the same features.

DATA VISUALIZATION

Preliminary analysis focused on describing the diversity of Tucson's wards by wealth index and household counts. Data cleaning addressed missing values and inconsistent formatting, and the wealth index was merged with ward-level identifiers to allow comparison across wards.

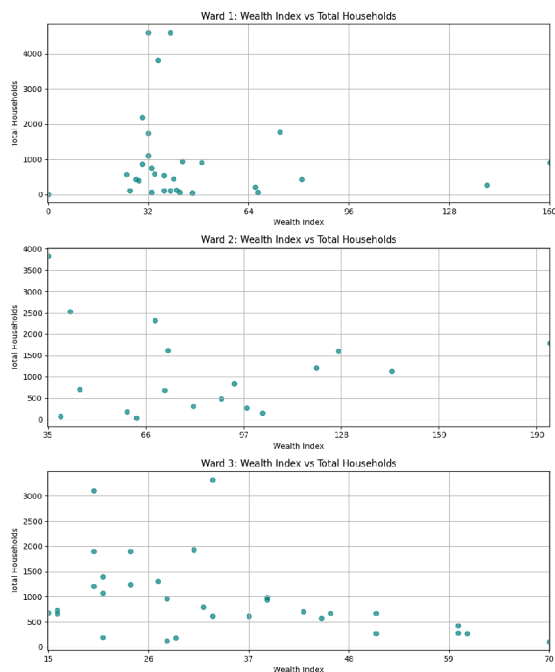


Figure 2 - Wealth indexes vs total houses, wards 1-3.

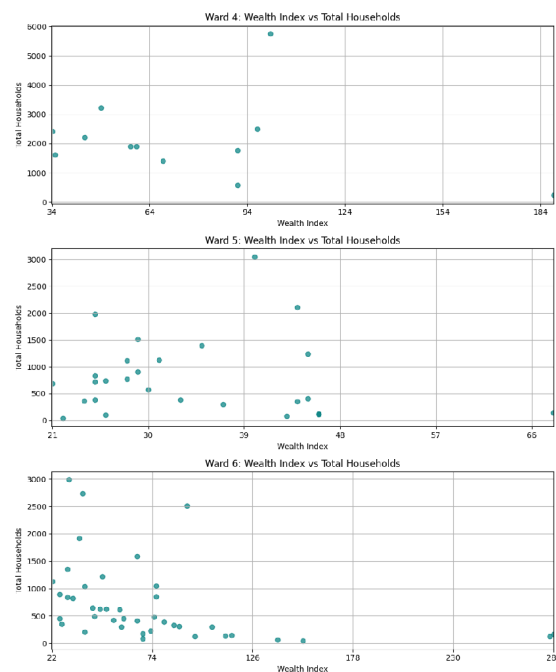


Figure 3 - Wealth indexes vs total houses, wards 4-6.

Ward 1 shows a concentration of lower wealth-index scores alongside a cluster of very high household counts, suggesting dense, lower-income housing. Ward 2 shows a wider, more mixed distribution of wealth index values, consistent with a more economically varied population. Ward 3 shows concentrated lower wealth-index scores across most neighborhoods, consistent with sustained economic disadvantage. Ward 4 shows consistently higher wealth-index scores with comparatively little spread, consistent with a more affluent, economically stable population. Ward 5 shows a narrow range of predominantly lower wealth-index scores. Ward 6 shows the widest spread of wealth-index values of any ward, indicating greater internal economic diversity.

The second part of this analysis examines the hourly distribution of crimes across Tucson's wards. This required cleaning raw crime records — extracting ward identifiers, converting reported timestamps into usable hourly values, and removing records with invalid or missing time data — to compute crime frequency by ward and hour of day.

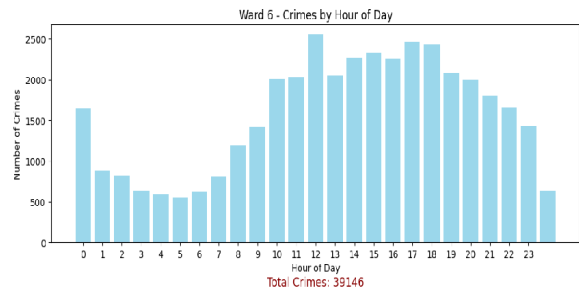
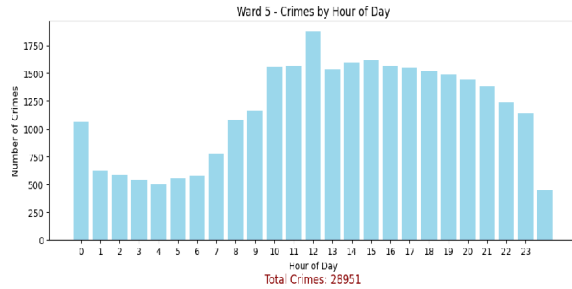
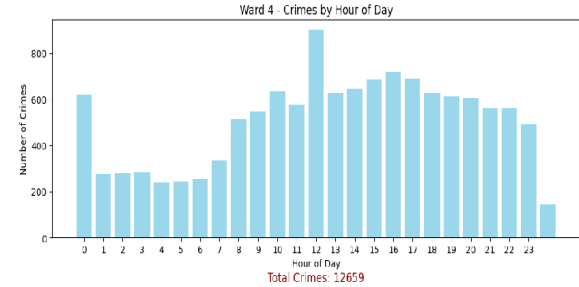
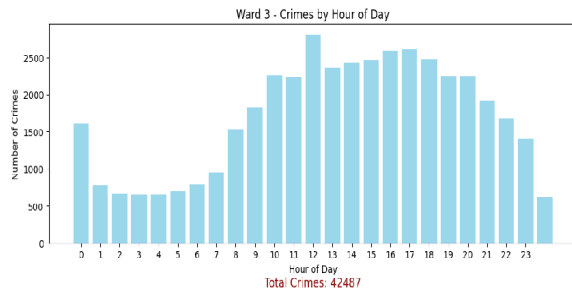
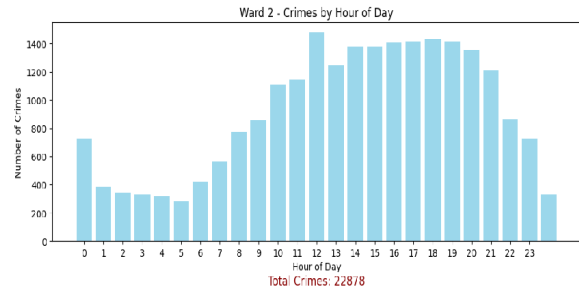
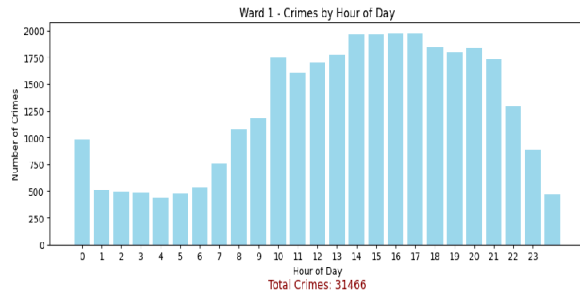


Figure 4 - Crimes vs hour of the day, wards 1, 3, 5.

Figure 5 - Crimes vs hour of the day, wards 2, 4, 6.

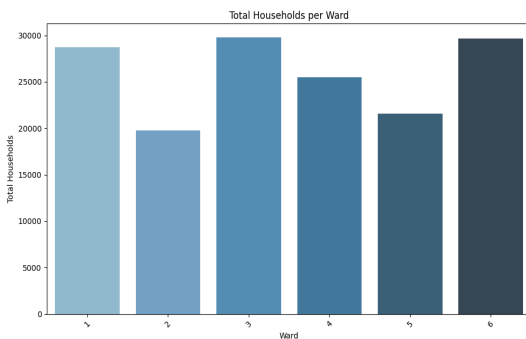


Figure 6 - Total houses per ward

Ward 2 and Ward 5 have noticeably fewer total households than the other four wards, with Ward 2 the lowest at roughly 19,700. Wards 1, 3, and 6 each have around 28,000–30,000 households, making them the most densely populated by household count. This variation matters for interpreting the crime and wealth figures, since wards with more households naturally have more opportunity for reported incidents independent of any economic effect.

Across all six wards, crime volume is lowest in the very early morning hours and rises through the late morning, peaking around midday into the early evening before declining overnight. This general pattern holds across wards regardless of total crime volume, suggesting it reflects citywide activity rhythms rather than a ward-specific effect. Total crime counts vary substantially

by ward, with Ward 3 recording the highest total (56,005) and Ward 4 the lowest (16,402) over the period examined.

Figure 7 (below) shows the distribution of crime types by ward. Larceny is the most common crime type in every ward by a wide margin, indicating that property crime of this kind is a citywide pattern rather than one concentrated in specific economic conditions. Other crime types — robbery, aggravated assault, and burglary — show more variation across wards, which is consistent with the possibility that some crime types relate more closely to neighborhood economic conditions than others, though this figure alone cannot establish that relationship.

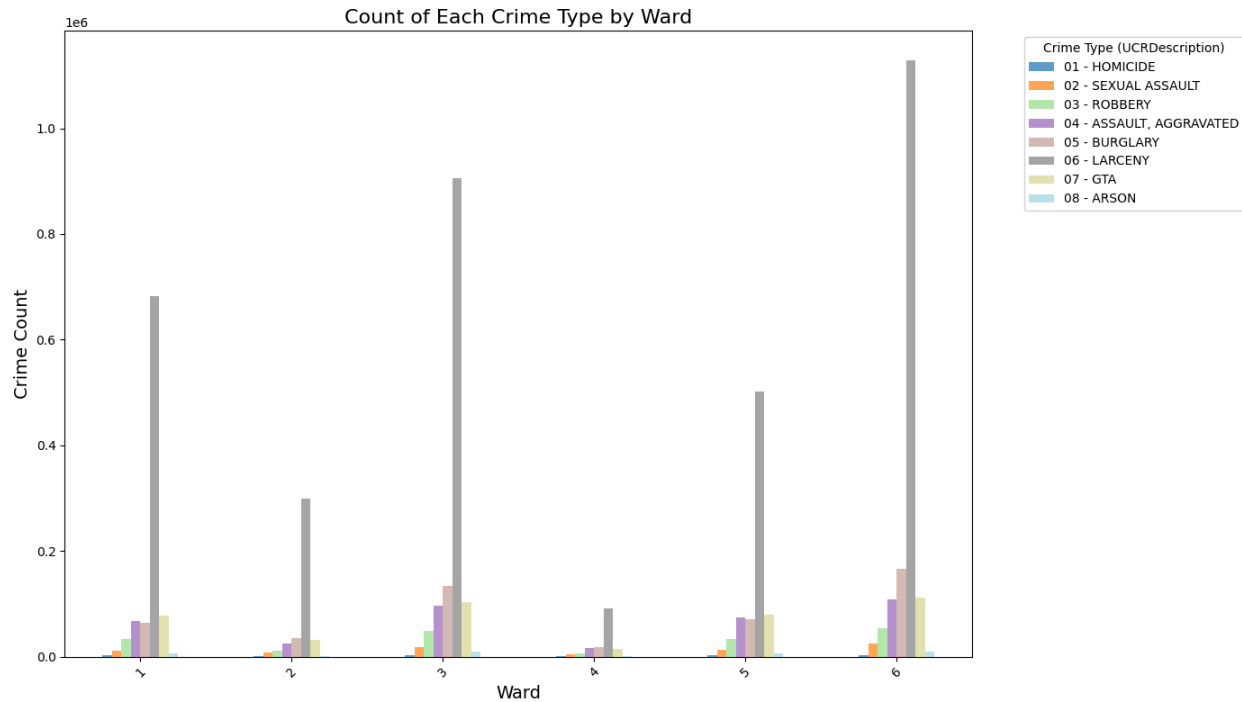


Figure 6 - Crime type and count, per ward.

PREDICTIVE MODEL 1 - LINEAR REGRESSION

The first model is a linear regression predicting a neighborhood's total crime count from two features: Wealth Index and Total Households. One high-leverage outlier neighborhood was excluded from training, consistent with the original analysis. The data was split 80/20 into training and test sets. The model produced a Wealth Index coefficient of -2.44 and a Total Households coefficient of 0.21 , indicating that, within this model, higher neighborhood wealth is associated with lower predicted crime counts, while a larger number of households is associated with modestly higher predicted crime counts — an intuitive result given that more households generally means more opportunity for reported incidents. The model achieved a mean squared error of 933,382, compared to a baseline (mean-only) MSE of 1,167,674, indicating the model captures some real signal beyond a naive average. The model's R^2 was 0.20, meaning it explains roughly one-fifth of the variance in neighborhood crime counts using income and household data alone. This is a modest but genuine relationship: a large majority of the variation in crime counts across neighborhoods is driven by factors outside this model — policing patterns, land use, population density, and other structural factors not captured in this dataset.

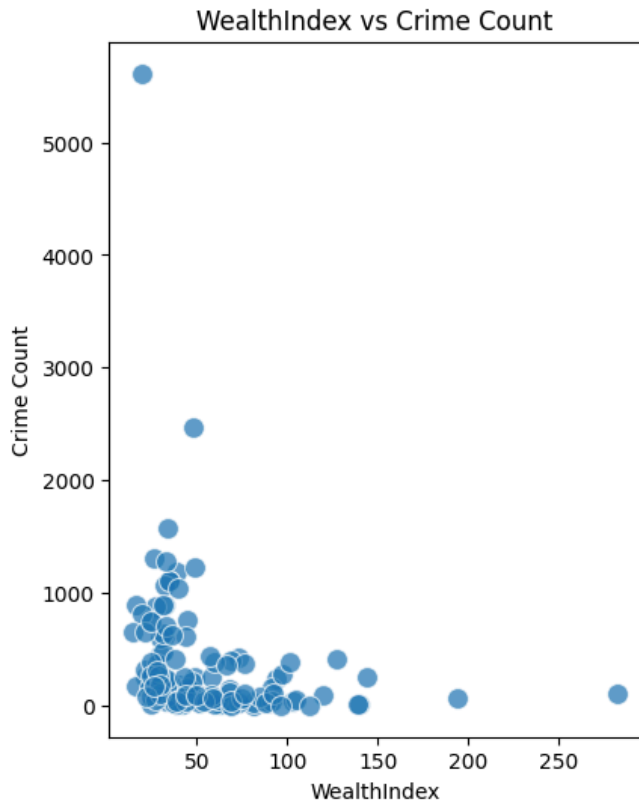


Figure 7 - Wealth vs. crime counts across wards.

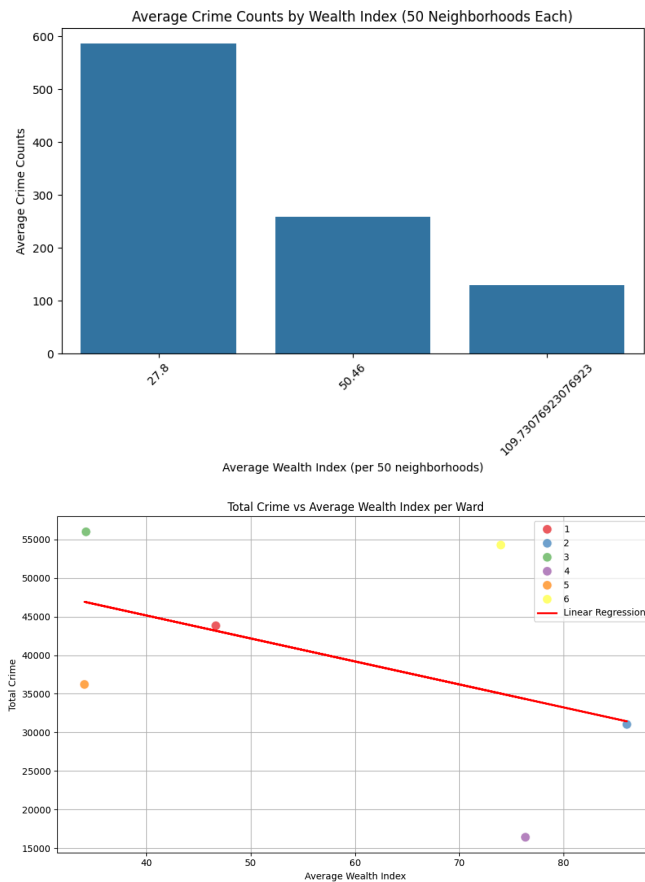


Figure 8 - Avg. crime vs wealth index.

Figure 9 - Total crime vs avg. wealth index.

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PREDICTIVE MODEL 1 - LOGISTIC REGRESSION

The second model reframes the task as binary classification: predicting whether a neighborhood falls above or below the median crime count (“high-crime” vs. “low-crime”), using the same two features — Wealth Index and Total Households — with 5-fold cross-validation to select regularization strength.

The model achieved 81% accuracy on the test set, compared to a 50% baseline (predicting the majority class). The confusion matrix correctly classified 11 of 13 low-crime neighborhoods and 10 of 13 high-crime neighborhoods, with precision, recall, and F1-scores each in the 0.79–0.85 range across both classes — a reasonably balanced result. The ROC curve produced an AUC of 0.93, indicating strong ability to rank neighborhoods by likelihood of falling into the high-crime category across classification thresholds.

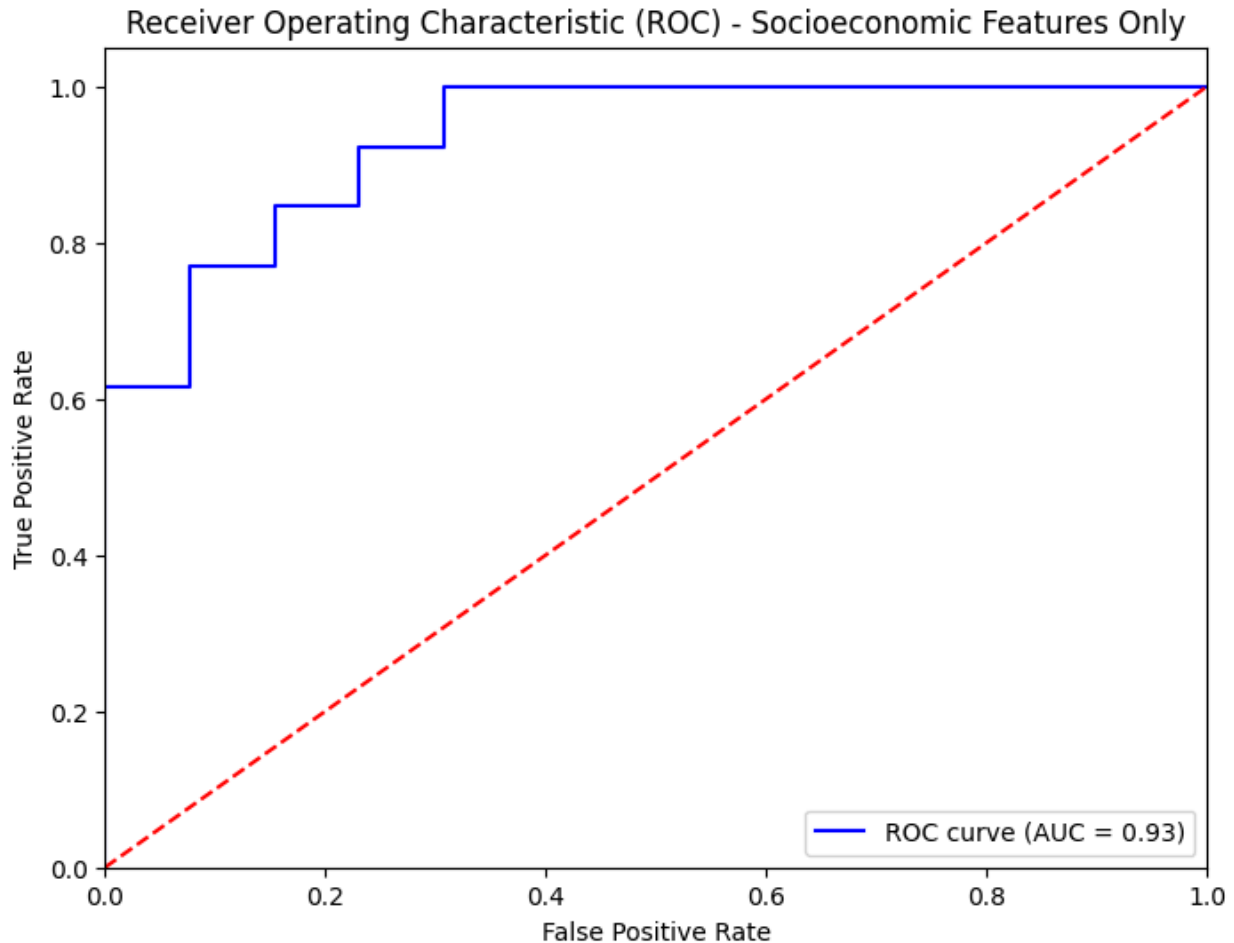


Figure 10 - ROC curve, logistic regression (income and household features only).

Results

This analysis found a consistent, moderate association between neighborhood income and crime rates in Tucson using only two socioeconomic features — Wealth Index and Total Households. Descriptive visualization showed that wards with lower average wealth index scores (Wards 1, 3, and 5) tend to show higher total crime counts, while wealthier, more economically stable wards (2 and 4) tend to show lower totals — consistent with the pattern described in the related works reviewed above. Temporal analysis showed a consistent citywide rhythm to crime volume, peaking from midday into early evening, independent of a ward's economic profile.

The two predictive models offer a useful contrast. The classification model (Model 2) performed strongly — 81% accuracy and an AUC of 0.93 — suggesting that income and household data alone are fairly effective at distinguishing higher-crime from lower-crime neighborhoods in broad terms. The regression

model (Model 1), by contrast, explained only about 20% of the variance in exact crime counts, indicating that while income and household data help identify which neighborhoods tend toward more or less crime, they are far less able to predict precisely how much crime a given neighborhood will experience. Together, these results suggest neighborhood economic conditions are a meaningful but partial explanation for differences in crime across Tucson's wards — not a complete one.

Several limitations are worth stating directly. First, this analysis uses neighborhood- and ward-level aggregated data; relationships observed at this level do not necessarily hold for individuals within those neighborhoods (the ecological fallacy), and no claim is made here about individual behavior. Second, crime data reflects reported and recorded incidents, which are shaped by policing patterns and reporting behavior as well as by underlying criminal activity — areas with more policing presence may show more recorded crime independent of the true underlying rate. Third, all relationships reported here are correlational; this analysis does not establish that income causes crime, or vice versa, and likely confounding variables (population density, urbanization, land use, historical policy) are not fully accounted for in a two-feature model. An earlier version of this analysis also included neighborhood race and ethnicity demographic data as model features; that data has been removed, as this project's hypothesis concerns economic conditions, not demographic composition, and using race as a predictive input for a crime-classification model risks implying a causal or predictive claim about demographic groups that this analysis was never designed to make and the data cannot responsibly support.

These findings are most appropriately read as pointing toward where economic investment and stability programs might have the most relevance — not as a tool for predicting or profiling individual neighborhoods or residents. Future work could extend this analysis with longitudinal data across multiple years, additional non-demographic socioeconomic variables (education access, housing stability, employment rates), and a more explicit discussion of the ethical considerations involved in any predictive approach to crime and policing.

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